

Bounds of Cycle Integrals of Modular Forms

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Background Information: Continued Fraction

Any real number can be written as:

$$w = [a_0; a_1, a_2, a_3, \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$$

$$\varphi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}} = [\overline{1}]$$

$$\pi = 3 + \frac{1}{7 + \frac{1}{15 + \frac{1}{1 + \frac{1}{292 + \dots}}}}$$

Key Fact: For quadratic irrationals $w = \frac{a+b\sqrt{D}}{c}$, the continued fraction is **eventually periodic**:

$$w = [a_0; a_1, a_2, \dots, a_m, \overline{a_{m+1}, \dots, a_m}]$$

Background Information: Matrix Action

We let a matrix A acts on a complex number w by a Möbius transformation:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \qquad A \cdot w = \frac{aw + b}{cw + d}$$

We associate to the matrix A a quadratic form: $Q(x) = Ax^2 + Bx + C$,

$$A = c, \quad B = d - a, \quad C = -b.$$

So that the fixed points of the matrix A are exactly the roots of this quadratic form

Introduction to Modular Forms: The Value Function

A **modular form** of weight k is a **holomorphic function** $f: H \rightarrow \mathbb{C}$ from the **upper half-plane** satisfying the following two conditions for any matrix $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$

- *Automorphy condition*: $f(\gamma(z)) = (cz + d)^k f(z)$;
- *Growth condition*: the function $(cz + d)^{-k} f(\gamma(z))$ is bounded for $\mathrm{im}(z) \rightarrow \infty$

Given a quadratic irrational w that is a fixed point of a hyperbolic matrix $A \in \mathrm{SL}_2(\mathbb{Z})$, we can associate to A such a quadratic form Q . Define the value of w under a modular form f as:

$$I_f(w) = \int_{\tau}^{A\tau} \frac{f(z) \sqrt{D}}{Q(z, 1)} dz \quad (\text{Note: "Hyperbolic" means } \mathrm{Re}(\lambda) \neq 0)$$

$$\mathrm{val}_f(w) = \frac{I_f(A)}{2 \log \varepsilon} \quad \varepsilon \text{ is the largest eigenvalue of } A$$

Interpretation:

Each quadratic irrational w is assigned a numerical “energy”

Problem Statement

$$j(\tau) = 1728 \frac{E_4(\tau)^3}{E_4(\tau)^3 - E_6(\tau)^2} = \frac{E_4^3(\tau)}{\Delta(\tau)}.$$

The modular j function is a function defined on the upper half-plane invariant under modular transformations

Kaneko's Conjecture

When $f = j$, and w is a quadratic irrational $\text{val}_f \left(\frac{1 + \sqrt{5}}{2} \right) \leq \text{Re}(\text{val}_f(w))$

When w is Markov irrational, the function also has a upper bound

Problem Statement

Problem 1:

Can the bound in the Kaneko's conjecture for the j -invariant be extended to modular forms $f \neq j$, especially at nonzero weight, e.g.

$$f = \frac{E_4^2}{\Delta}$$

Problem 2:

Find all $P(x) \in \mathbb{R}[x]$ such that the minimum value of $\operatorname{Re}(\operatorname{val}_{P(j)}(w))$ is $\operatorname{Re}(\operatorname{val}_{P(j)}(\phi))$.

Meaning: we replace j by a polynomial $P(j)$ and check whether the minimum still occurs at ϕ

Data Obtained So Far

Approach & Goal

Empirical Step: use SageMath to empirically determine possible bounds for various modular forms

Theoretical Goal: Based on these computational results, adapt the existing proof techniques to formally prove the lower bound.

Empirical Results: Just as with the j -function, we found that for $f = \frac{E_4^2}{\Delta}$, the inequality $\text{val}_f\left(\frac{1+\sqrt{5}}{2}\right) \leq \text{Re}(\text{val}_f(w))$ continues to hold for nearly 15,000 matrices of the form $A = T^{a_1} V^{a_2} \dots T^{a_{2\ell-1}} V^{a_{2\ell}}$ with $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

i.e. the minimum is consistently attained at the golden ratio.

Data Obtained So Far

The minimum is consistently attained at the matrix TVTVTVTVTV, whose fixed point is the golden ratio.

a_list	matrix_A_abcd	trace	discriminant_D	val_f
(1, 1, 1, 1, 1, 1, 1, 1, 1, 1)	(89, 55, 55, 34)	123	15125	-99.92938911747913
(1, 1, 2, 2, 3, 1, 1, 2, 1, 2)	(911, 335, 533, 196)	1107	1225445	-51.833261726468635
(1, 1, 3, 1, 2, 3, 1, 2, 2, 2)	(1728, 713, 967, 399)	2127	4524125	-42.2228903936505
(1, 2, 1, 1, 2, 3, 3, 3, 3, 1)	(2857, 2193, 2061, 1582)	4439	19704717	-30.321167714055484
(1, 2, 2, 3, 3, 2, 1, 2, 2, 1)	(2373, 1669, 1682, 1183)	3556	12645132	-38.22551839417993
(1, 3, 1, 1, 2, 1, 3, 1, 1, 2)	(691, 270, 540, 211)	902	813600	-50.13709228742196
(1, 3, 2, 1, 3, 2, 3, 1, 3, 3)	(5965, 1827, 4597, 1408)	7373	54361125	-26.059903097162962
(1, 3, 3, 1, 1, 3, 1, 1, 1, 2)	(1006, 381, 771, 292)	1298	1684800	-43.97821730596991

Another Problem: The Polynomial Problem

Problem 2:

Find all $P(x) \in \mathbb{R}[x]$ such that the minimum value of $\operatorname{Re}(\operatorname{val}_{P(j)}(w))$ is $\operatorname{Re}(\operatorname{val}_{P(j)}(\phi))$.

We have already developed a SageMath program to calculate $I_f(w) = \int_{\tau}^{A\tau} \frac{f(z)\sqrt{D}}{Q(z, 1)} dz$ when $f = j$

Once we identify candidates for $P(x)$, we can easily modify this program to compute the integration value for $f = P(j)$